

RADIONUCLIDE SAFETY DATA SHEET					
RADIONUCLIDE: Cd-109			FORMS: Soluble		
PHYSICAL CHARACTERISTICS HALF-LIFE: 461.4 days DECAY EMISSIONS					
Gammas / X-rays		Betas / Positrons (+) / Electrons*		Alphas	
E (keV)	%	E (keV, Ave)	%	E (keV)	%
22	84	85*	43		
25	17	63*	41		
88	4	19*	14		
		88*	9		
- Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.					
STANFORD HAZARD CATEGORY C – level (low hazard): ≤ 200 µCi B – level (moderate hazard): > 200 µCi, ≤ 10 mCi A – level (high hazard): > 10 mCi					
EXTERNAL RADIATION HAZARDS Gamma dose rate, point source at 1 ft, 1 mCi: 0.028 mrem/h Beta dose rate to skin, point source at 1 ft, 1 mCi: 0 mrem/h Contamination skin dose, uniform deposit of 1 µCi per cm²: 2000 mrem/h			INTERNAL RADIATION HAZARDS Annual Limit on Intake: 300 µCi (Ingestion) 40 µCi (Inhalation) The values above indicate the activity taken into the body that would result in either 5 rem to the whole body (CEDE) or 50 rem to an organ or tissue (CDE).		
SHIELDING Gammas/X-rays: <1 mm of lead will reduce the gamma dose rate by 90%. Betas/electrons: <1 mm of plastic will absorb all emissions.			DOSIMETRY AND BIOASSAY REQS Urine assays may be required after large spills or contaminations.		
SPECIAL PROBLEMS AND PRECAUTIONS: 1. Recommended survey probe: Thin-window NaI 2. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often. 3. Survey work areas before, during, and after work. Instrument and smear surveys are required. 4. Segregate waste to those with half-lives of greater than 120 days (excluding H3 and/or C14). 5. Limit soluble waste to the sewer to less than 1 µCi/day per lab.					

References:

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- ICRP, 2008. Nuclear Decay Data for Dosimetric Calculations. ICRP Publication 107. Ann. ICRP 38 (3).
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- Smith, D., Stabin, M. (2012) Exposure Rate Constants and Lead Shielding Values for Over 1,100 Radionuclides. *Health Physics*, 102(3): 271-291.
- 10.CFR.20 – Standards for Protection Against Radiation (2019). Retrieved from <https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/>